INF 2002 HUMAN COMPUTER INTERACTION

Final Report

Scenario Based Fire Preparedness Game

Team Number: 21

Team Name: Team T

Due Date: 27 November 2025

NAME	**STUDENT ID**	**GITHUB HANDLE**
KWEK LI XUAN	2402517	ahhlxlx
SUBHASHINI RAJKUMAR	2402769	subhaa2
MUHAMMAD ADLI BIN JUMAHAT	2401516	adliiiiiiiiiiiiiiiiiiiiSIT
LAI YUEYIN SHYANN	2402456	shyannlaii
WONG JUN YONG BRIAN	2401610	Brian1z3

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Introduction

Our project aims to improve fire emergency preparation through an interactive, scenario based learning game. The game places users in realistic environments including kitchen and car, where they have to make decisions during simulated fire emergencies. These activities were chosen because fire response requires fast decision-making and scenario based learning enables users to practice key actions, such as recognising risks, deciding evacuation routes or selecting the correct tool to put out fire in a safe and controlled environment. This is in line with the design concept of “Simple and Interactive Interfaces” by providing clear and intuitive interactions that guide users through realistic safety scenarios. To ensure that the interface effectively encourages learning, we conducted usability tests to determine how well users interpret key buttons and interface feedback, allowing us to continuously enhance clarity, responsiveness and overall user experience.

Objectives and Hypotheses

Objectives (PO)

ID	Description
PO1	Fix the issue of the “Scenario Info” button not being understood by most testers, increasing the clarity of its purpose.
PO2	Improve the perceived feedback and responsiveness of button interactions by adding sound effects, thereby increasing overall user satisfaction with the interface.

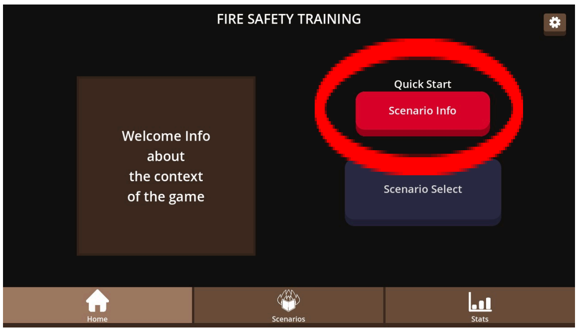
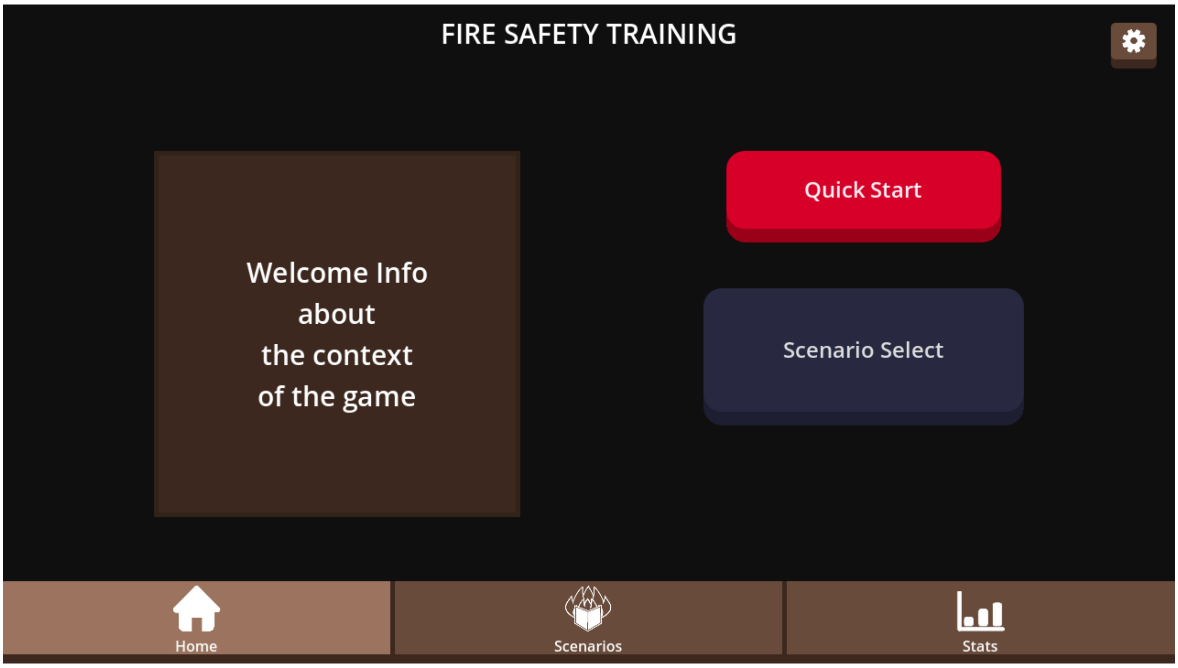
Hypothesis (PH)

ID	Description
PH1	Changing the button label from “Scenario Info” to “Quick Start” will increase the clarity of the button’s purpose.
PH2	Participants who use the interface with button sound effects will report higher satisfaction with the button feedback than participants who use the interface without sound effects.

Change Record

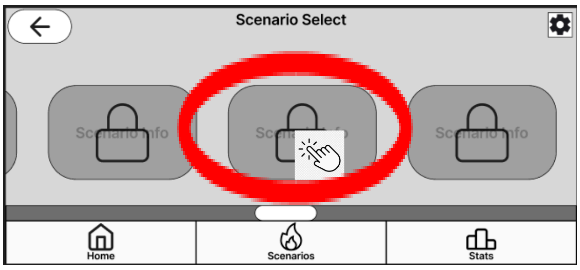
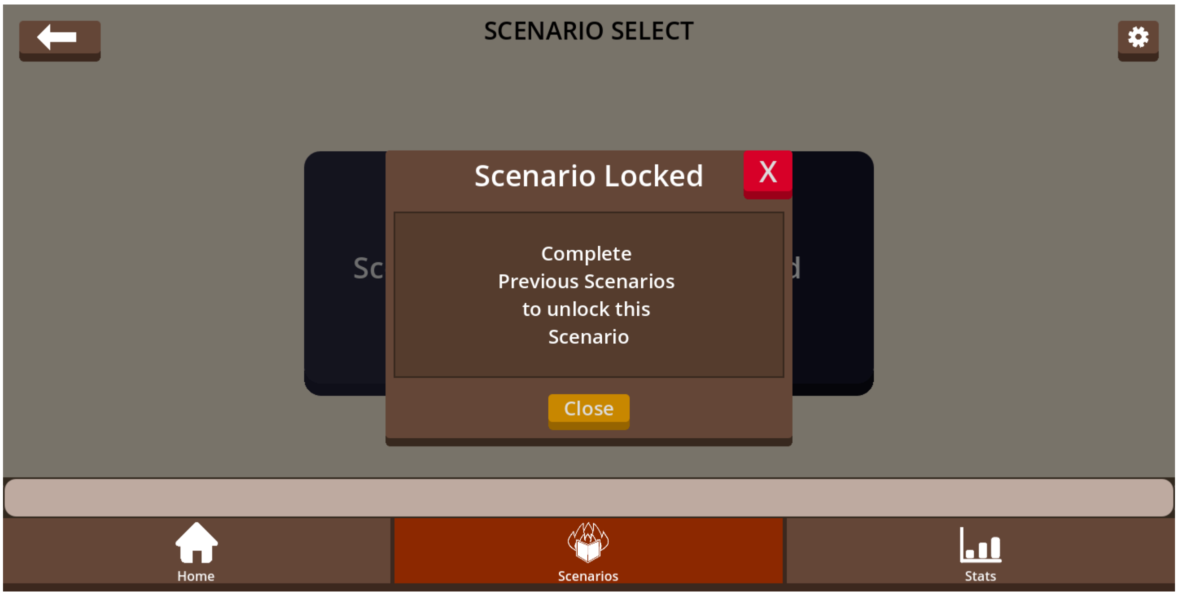
Change 1

Screen ID: Home

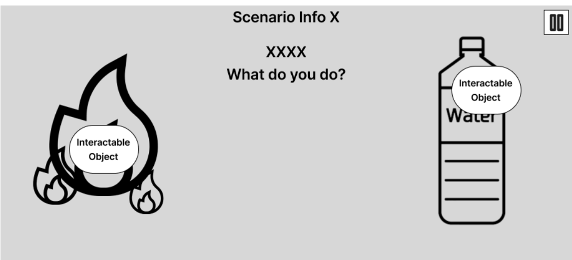
<table>
<tr><td colspan="2">Title: Misleading Button Label</td></tr>
<tr><td>Screenshot of the issue(s) with the object

</td><td>Description of the issue(s) with the object:
The “Scenario Info” button was meant to bring the user to their last played scenario. However, users thought the button would just give information on the current scenario.</td></tr>
<tr><td colspan="2">Changes:
1. Changed the button label from “Scenario Info” to “Quick Start” to better match common game UI conventions for quickly starting/continuing gameplay.

</td></tr>
<tr><td>Date of change: 14/11/25</td><td>Version: 2</td></tr>
</table>

Change 2

<table>
<tr><td colspan="2">Screen ID: Scenario Select</td></tr>
<tr><td colspan="2">Title: No response when tapping on locked scenarios</td></tr>
<tr><td>Screenshot of the issue(s) with the object</td><td>Description of the issue(s) with the object:
Tapping on a locked scenario does not show any response to the user. The user may remain confused on how to unlock that scenario.</td></tr>
</table>

<table>
<tr><td>
</td><td></td></tr>
<tr><td colspan="2">Changes:
1. Add a popup that appears when a user clicks on a locked scenario.
2. The popup should contain information on how to unlock the locked scenario.
3. The background of the popup should have a black translucent color to indicate to the user that they need to close the popup first to proceed.

</td></tr>
<tr><td>Date of change: 14/11/25</td><td>Version: 2</td></tr>
</table>

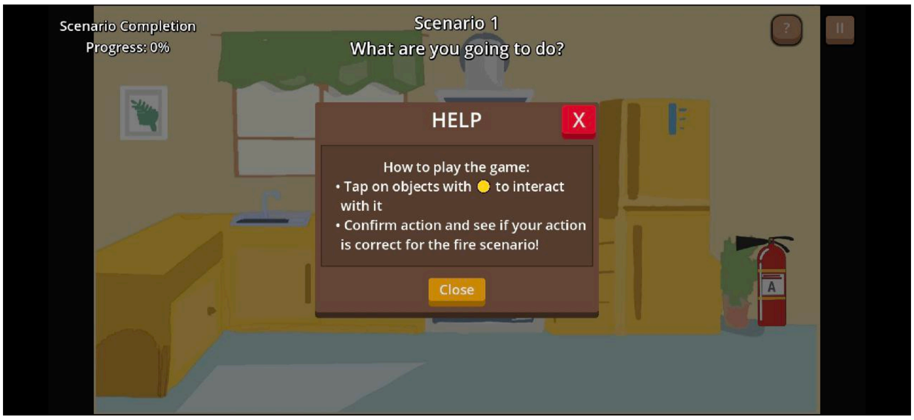
Change 3

<table>
<tr><td colspan="2">Screen ID: Scenario</td></tr>
<tr><td colspan="2">Title: Lack of instructions on how to play</td></tr>
<tr><td>Screenshot of the issue(s) with the object

</td><td>Description of the issue(s):
Users were unsure of how to interact with the objects as there were no instructions of how to play the game.</td></tr>
</table>

Changes:

1. Added a “help” button next to the pause button in the scenario screens. This provides users with a clear, dedicated place to access guidance.

2. The “help” button will pause the game and display a pop-up containing the instructions on how to play the game.



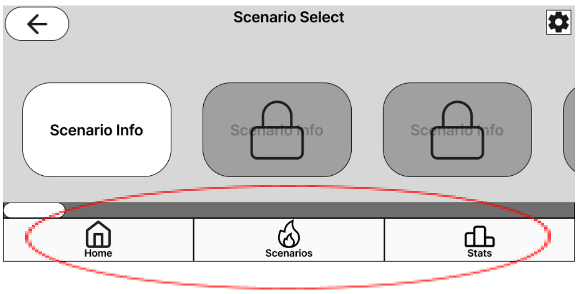
Date of change: 22/11/25	Version: 3 (Final)

Change 4

Screen ID: Scenario Select, Home, Scenario Statistics

Title: Tab was not highlighted on the navigation bar.

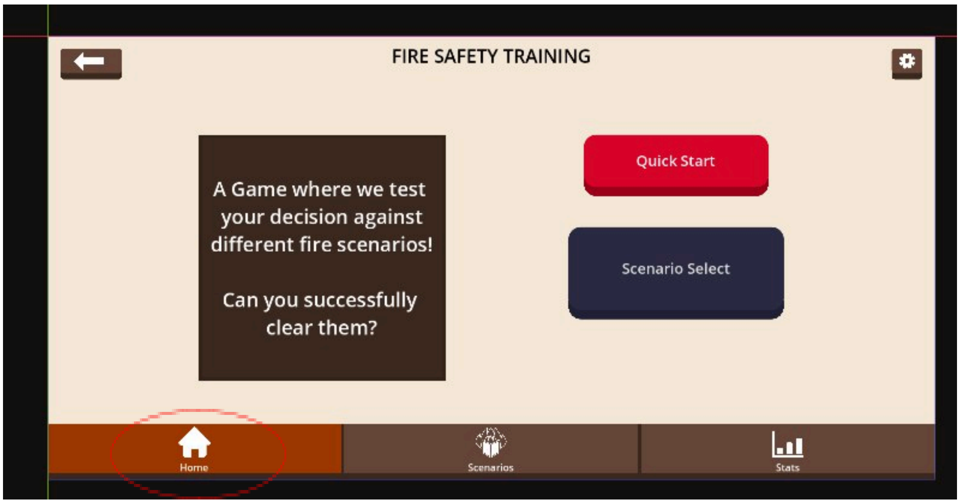
Screenshot of the issue(s) with the object



Description of the issue(s):
The bottom navigation bar did not indicate the active tab, so users could not tell which screen they were currently on.

Changes:

1. Added an active tab indicator on the bottom navigation bar to show which screen the user is currently on.



Date of change: 15/11/25	Version: 2

Change 5

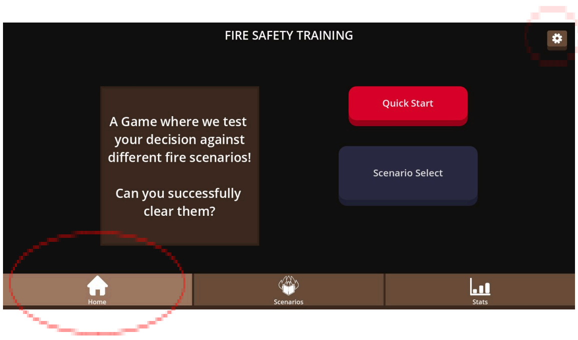
Screen ID: Scenario Select, Home, Scenario Statistics

Title: Colour theme

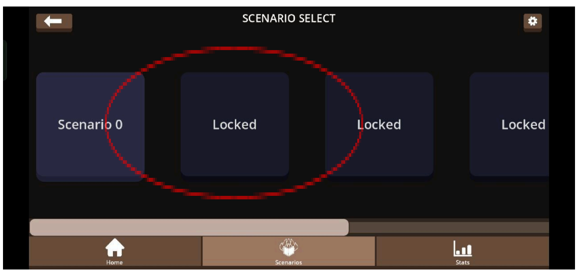
Screenshot of the issue(s) with the object	Description of the issue(s):

Screenshot of the issue(s) with the object

Home screen:



Scenario Select screen:



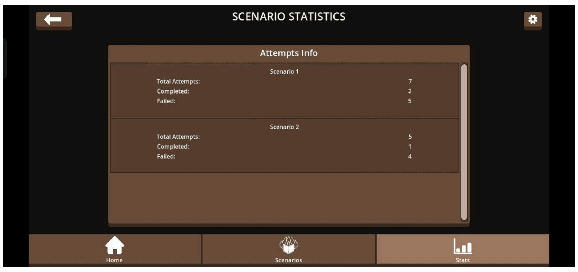
Scenario Statistics screen:

Description of the issue(s):
Some of the colors used in the text color to button color or button color to background color does not have a good contrast ratio and did not reach the minimum color contrast ratio of 4.5

Eg. Highlighted tab color (#9C7762) and the icon color (#FFFFFF) has a color contrast of 4.02

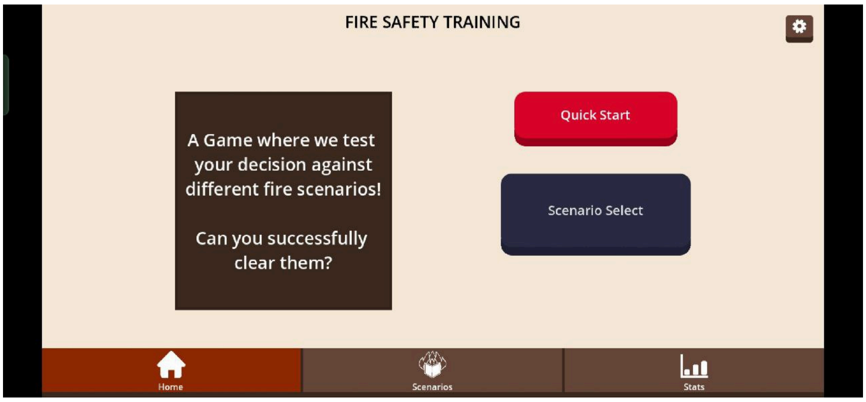
Eg. Setting Button color (#6A4E3C) and the background color (#131010) has a color contrast of 2.5

Eg. Scenario Locked Button color (#1A1B2B) and background color (#131010) has a color contrast of 1.11

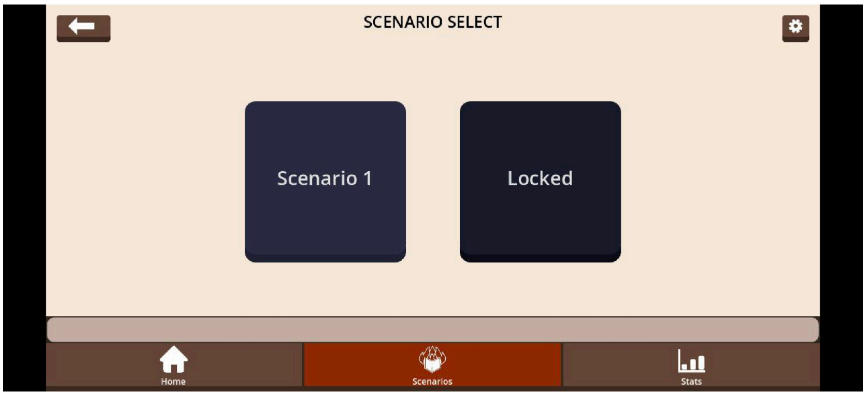


Changes:

1. Change the background colour of the screens, navigation bar and some buttons.
    a. Updated Home screen:
    - Background colour change from #131010 to #F5E8D7 which increases the contrast ratio between the background and the brown color buttons (like Setting button and Navigation Buttons) from 2.5 to 7.
    - Navigation bar highlighted tab colour change from #9C7762 to #8F2903 which increases the contrast ratio between the background and the button color from 4.02 to 7.
    - Navigation bar highlighted tab color change from #9C7762 to #8F2903 which increases the contrast ratio between the navigation icon and the navigation button from 4.02 to 8.45.



    b. Updated Scenario Select screen:
    - With the updated background color (#F5E8D7), the contrast ratio between the locked scenario button and the background increased from 1.11 to 14.09.



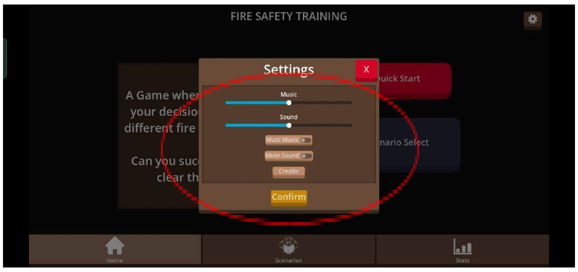
c. Updated Statistics screen:

Date of change: 26/11/25	Version: 3 (final)

Change 6

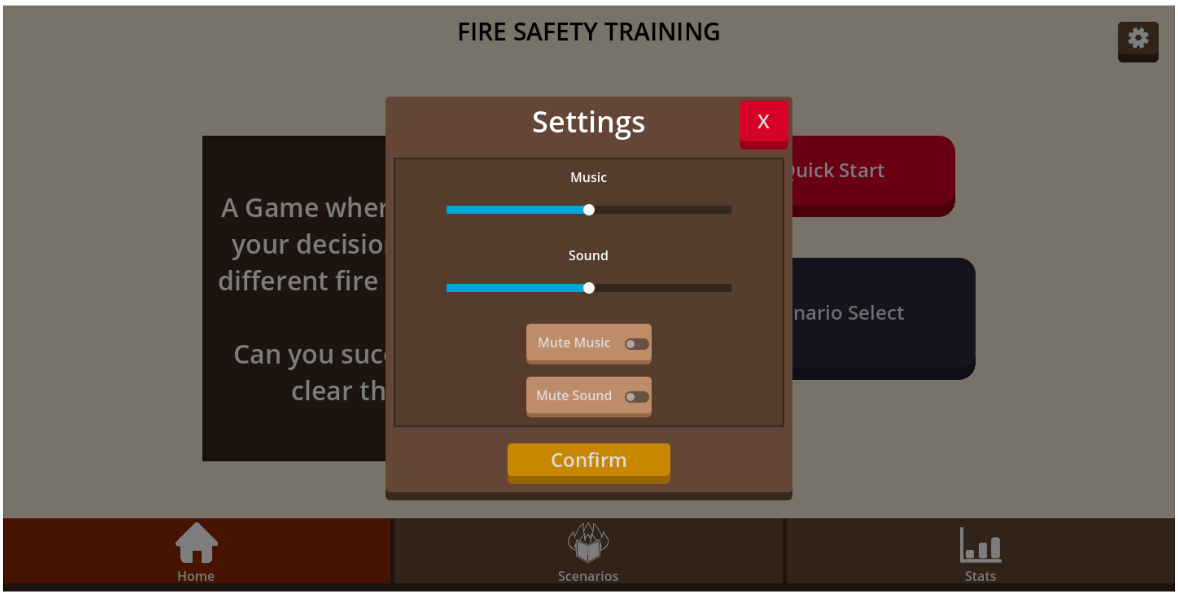
Screen ID: Settings Popup

Title: Inconvenient to tap / interact with setting options

Screenshot of the issue(s) with the object	Description of the issue(s):
	Buttons and slider options in the setting menu have small hitbox and is hard to interact with

Changes:

1. Updated Sliders and buttons size in Setting popup

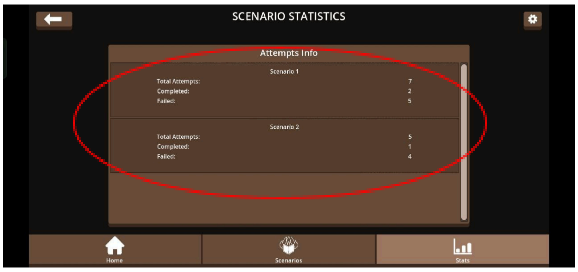


Date of change: 26/11/25	Version: 3 (Final)

Change 7

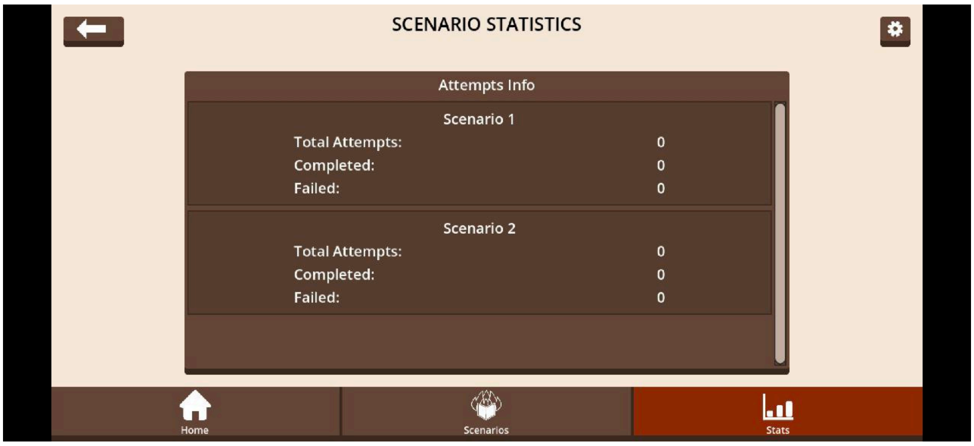
Screen ID: Scenario Statistics

Title: Text font size too small

Screenshot of the issue(s) with the object	Description of the issue(s):
	The text on the statistics screen was too small which made it hard to read on a mobile device.

Changes:

1. Updated Statistic Text size from 16px to 24px



Date of change: 26/11/25	Version: 3 (Final)

Change 8

Screen ID: All screens

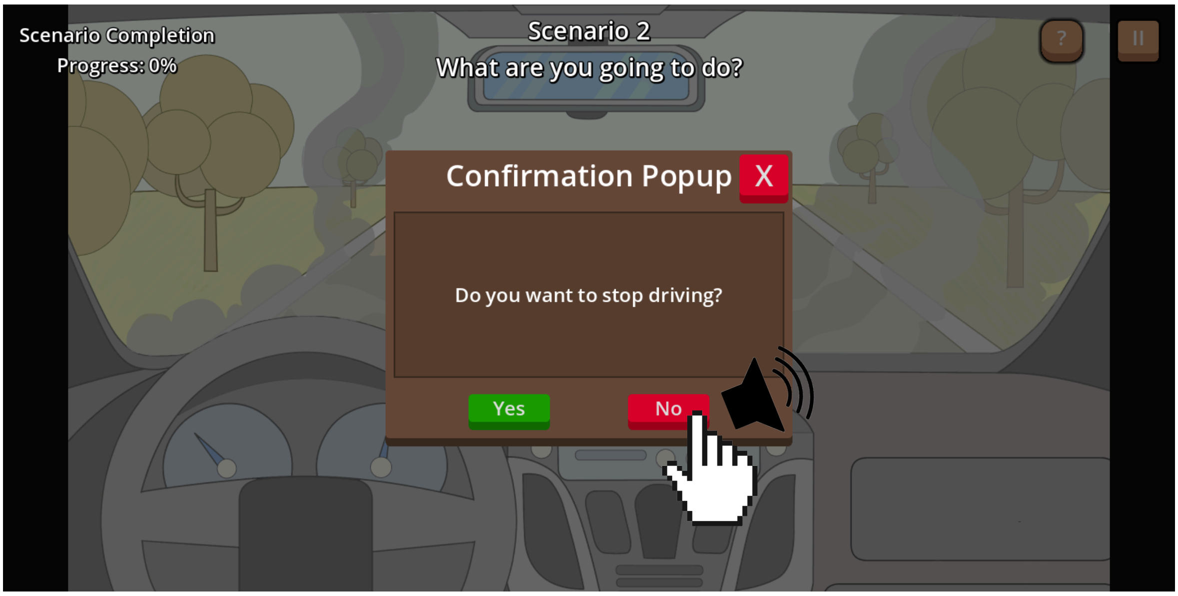
Title: Sound Effects (SFX) on button press

Screenshot of the issue(s) with the object	Description of the issue(s):
	Button presses lacked clear feedback, making interactions feel unresponsive.

Changes:

1. Added sound effects (SFX) to all button presses to increase the
    a. The game will now alternates between 2 variation of the “select” sfx sound whenever the player press a button

b. A distinct “no” SFX will be played whenever the player declines a confirmation prompt



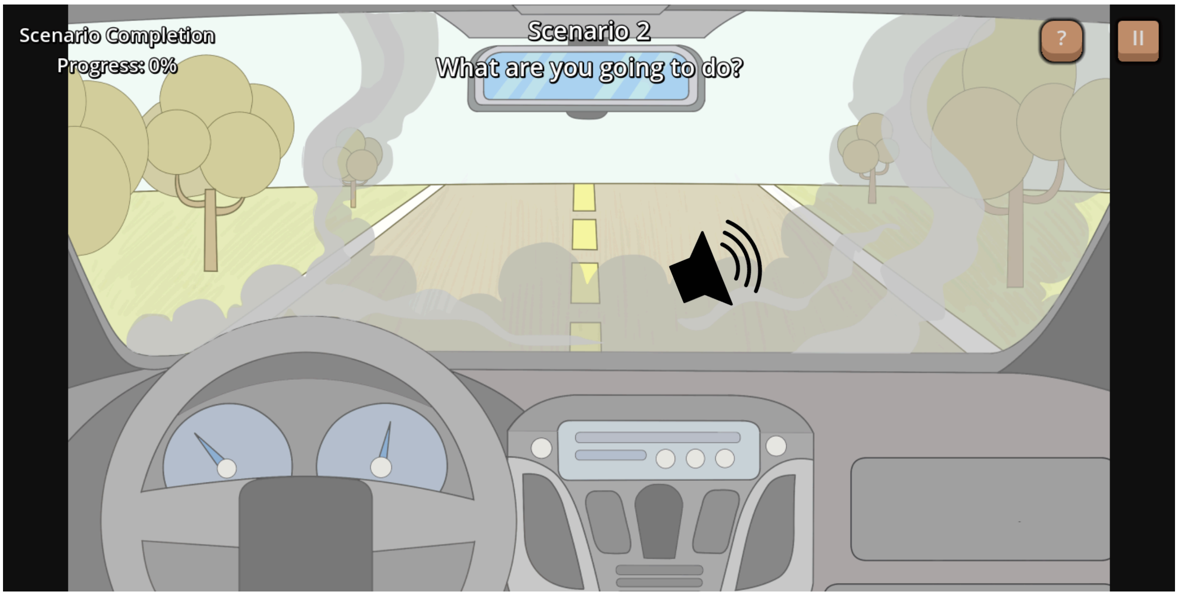
Date of change: 25/11/25	Version: 3 (Final)

Change 9

Screen ID: Scenario	
Title: Lack of audio in scenarios	
Screenshot of the issue(s) with the object	Description of the issue(s): The previous prototype had no background audio or sound effects, which reduced immersion and made the scenarios feel less

	engaging and atmospheric during gameplay.

Changes:
1. Added background music to the game
2. Added various of sfx of the different interactive objects in different scenario
    a. Eg. Add a car brake sound sfx to be played when the user choose to stop driving in the car fire scenario
    b. Eg. Add a car engine running sfx while the user have yet to turn off the car engine in the car fire scenario
    c. Eg. Add a gas running sfx while the user have yet to turn off the stove in the kitchen fire scenario



Date of change: 25/11/25	Version: 3 (Final)

Change 10

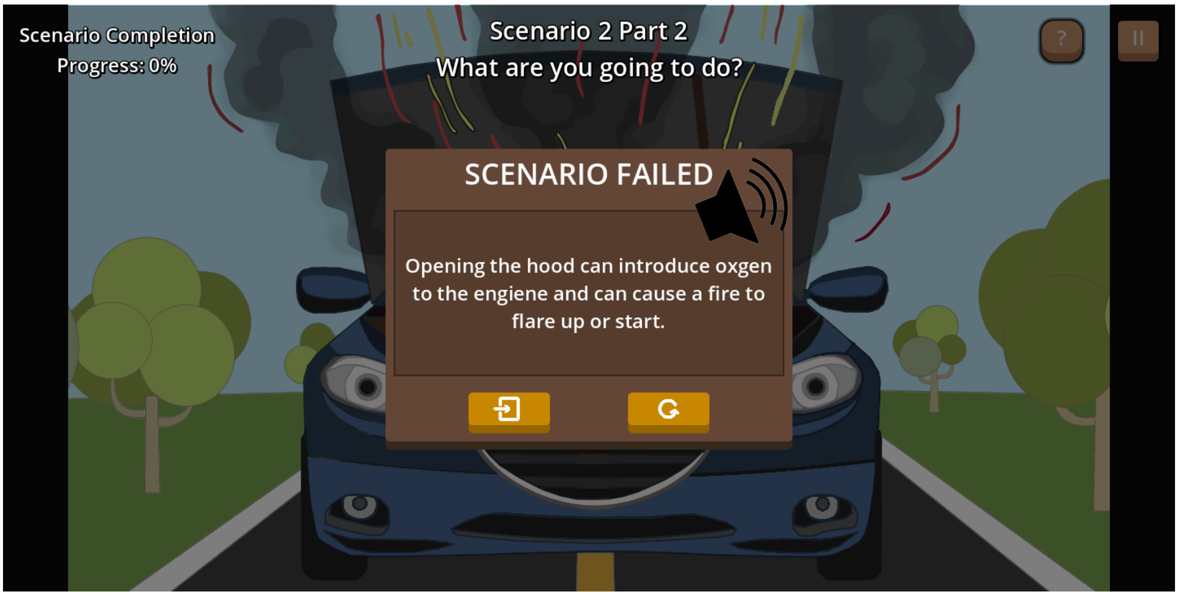
Screen ID: Scenario
Title: Added SFX when player win and lose

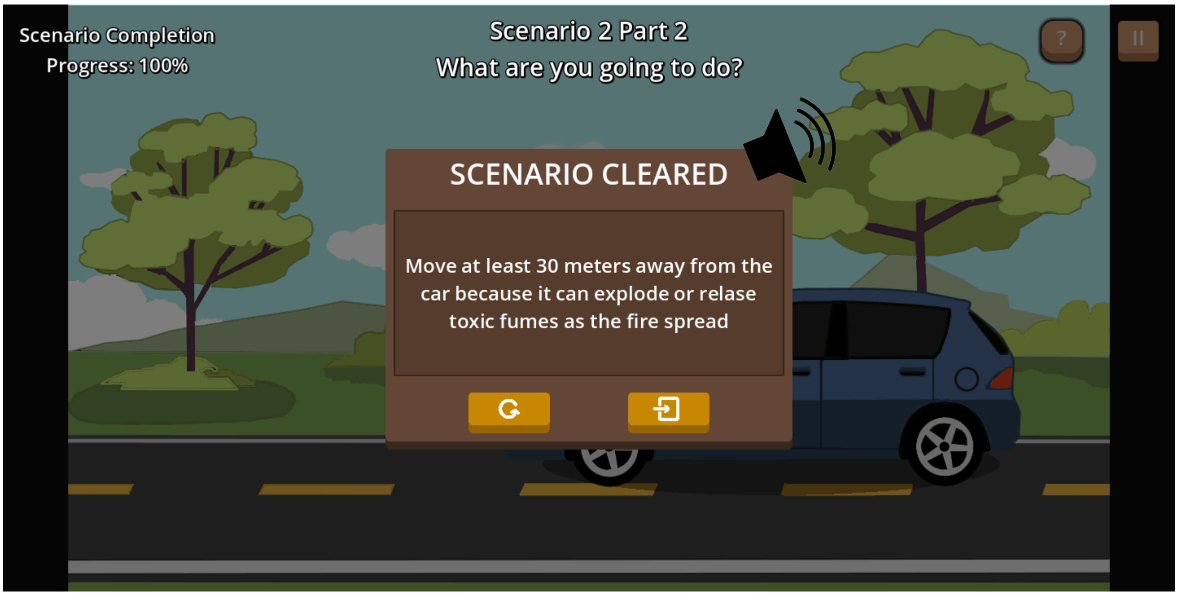
Screenshot of the issue(s) with the object	Description of the issue(s): The game lacked audio feedback for player actions, so interactions felt less responsive and players had less reinforcement of whether their choice was right or wrong, even though visual feedback was provided.

Changes:
1. Added SFX when player triggers win or lose conditions

a. When the user triggers a scenario failed, a “failed” sfx will be played along with the scenario failed popup to reinforce to the player that they performed a wrong choice

b. When the user trigger a scenario cleared, a “cleared” sfx will be played along with the scenario cleared popup to reinforce to the player that they performed a correct choice





Date of change: 25/11/25	Version: 3 (Final)

Experiments

Experiment 1: Button Label Clarity Test

Objective (PO1):

- Fix the issue of the “Scenario Info” button not being understood by most testers, increasing the clarity of its purpose.

Hypothesis (PH1):

- Changing the button label from “Scenario Info” to “Quick Start” will increase the clarity of the button’s purpose.

Experiment design

We will use a between-subjects design. Participants are randomly assigned to one of two conditions and see only one version of the main menu:

- Condition A: shortcut button labelled “Scenario Info”.
- Condition B: shortcut button labelled “Quick Start”.

Apart from this label, both versions are identical in layout, colours, and navigation, including an alternative longer path to access levels. We plan to recruit 10 testers, 5 per condition.

Task Given to Users

Participants are shown the main menu wireframe and must reach a playable scenario/level. The screen contains one main shortcut button (“Scenario Info” or “Quick Start”, depending on condition) and one alternative path that requires several extra steps.

Instructions: ”Imagine you are returning to this game and want to access your last played level. Using this interface, try to navigate to your last played level.”

Full script at Appendix A.

Variables and Data

Independent Variables (IVs):

- Button label (“Scenario Info” vs “Quick Start”)

Dependent Variables (DVs):

- Task success – used the tested button to reach a level: (Yes/No).
- Task completion time – time in seconds from menu display to level screen.
- Number of misclicks.
- Perceived clarity – “How clear was this button’s purpose?” (1–5 scale).

Control Variables:

- Same UI layout, navigation, device type, environment and navigation flow.
- Standardised task instructions.
- Participants have no prior exposure to this prototype.

Data types:

- Nominal: Task success (Yes/No).
- Ratio: Completion time (seconds), number of misclicks (counts).
- Ordinal: Clarity rating (1–5 scale).

Data collection:

- Observer times the task and records button use and misclicks.
- Post-task questionnaire to capture the clarity rating.

Experiment 2: Button Feedback Satisfaction Test

Objective (P02):

- Improve the perceived feedback and responsiveness of button interactions by adding sound effects, thereby increasing overall user satisfaction with the interface.

Hypothesis (PH2):

- Participants who use the interface with button sound effects will report higher satisfaction with the button feedback than participants who use the interface without sound effects.

Experiment design

We use a between-subjects design. Participants are randomly assigned to one of two conditions and experience only one version of the interface:

- Condition A: buttons play a short sound effect (SFX) on press.
- Condition B: buttons are silent and only provide visual feedback.

Both versions are otherwise identical in layout, colours and navigation. We plan to recruit 10 testers, 5 per condition.

Task Given to Users

Participants are shown the main menu wireframe and asked to complete a short fixed flow so they press several buttons:

1. Quick start a scenario from the main menu.
2. Return to the main menu.
3. Open the settings menu.
4. Go back to the main menu again.

Instructions: "Using this interface, please complete the steps shown on screen."

Full script at Appendix B.

Variables and Data

Independent Variables (IVs):

- Presence of button SFX (with SFX vs without SFX)

Dependent Variables (DVs):

- Satisfaction rating with button feedback: "Overall, how satisfied were you with the button feedback in this interface?" (1–5 scale)

Control Variables:

- Same UI layout, navigation, device type, environment and task flow for all participants.
- Standardised instructions that do not mention sound effects.
- Participants have no prior exposure to this prototype.

Data types:

- Ordinal: Satisfaction rating (1–5 scale).

Data collection:

- Post-task questionnaire to capture the satisfaction rating.

Result and Discussions

- Present your experiment results and explain what do the result mean. Analyse the results to form your conclusion on how well the individual objectives (P0x) and hypotheses (PHx) are met.

Experiment 1

Condition A (button labelled "Scenario Info") results:

Use Tested Button	Time Taken to get to Scenario Screen	No. of Misclicks	Clarity Rating (1 - 5) 1 = Very unclear 5 = Very clear
No	12.62	3	1
No	15.45	5	2
No	9.36	2	1
No	20.41	7	1
No	16.61	4	1

Condition B (button labelled "Quick Start") results:

Use Tested Button	Time Taken to get to Scenario Screen	No. of Misclicks	Clarity Rating (1 - 5) 1 = Very unclear 5 = Very clear
Yes	5.21	0	5
Yes	4.13	0	5
Yes	6.12	0	5
Yes	5.78	0	5
Yes	4.78	0	4

With the result from the tables above, participants from group 40 who participated in condition A took longer to reach the scenario screen, with timings ranging from 9.36s to 21.41s and an average of 14-15 seconds. They also had a lot of misclicks with some participants recording 3-7 of them. The majority of the clarity ratings (1-2) indicated lack of understanding of the button's purpose.

In comparison, participants from group 42 who participated in condition B, completed the assignment much quicker, mostly between 4-6 seconds and with nearly no misclicks. Their clarity ratings were consistently 4-5, indicating that people understand what the button does.

Overall, the results show a significant difference between the two button labels in terms of task performance, misclick rate and perceived clarity. The "Scenario Info" button resulted in confusion, longer search time and more guessing before users found the correct button. Meanwhile, the "Quick Start" button resulted in significantly faster task completion and almost no mistakes, showing that a clear action-oriented label reduces confusion.

To determine whether the differences between the two conditions are statistically significant, we used non-parametric tests. This is because the sample size is small and the clarity rating is measured using a Likert scale (1–5), which is ordinal. We used Mann–Whitney U (two-tailed) to compare completion time, misclicks, and clarity rating between the two independent groups, and Fisher's exact test for the binary outcome (whether the participant used the tested button). Using $\alpha = 0.05$, the results were statistically significant. Button usage (0/5 vs 5/5) $p \approx 0.0079$. Completion time $U = 25.0$, $p \approx 0.0079$. Misclicks $U = 25.0$, $p \approx 0.0079$, Clarity rating $U = 0.0$, $p \approx 0.0079$. Overall, this supports PO1 and PH1.

In conclusion, both PO1 and PH1 are successfully met and supported. Participants using the "Quick Start" labeled button performed better in terms of speed, accuracy and clarity. This experiment. Thus, this experiment offers compelling evidence that action-oriented, simpler labels greatly enhanced usability.

Experiment 2

<u>Satisfaction Rating (1–5, 1 = Very dissatisfied, 5 = Very satisfied):</u>

Condition A: With Sound Effects	Condition B: Without Sound Effects
4	2
4	3
5	4
5	2
5	3
5	4

Based on the satisfaction ratings collected and presented in the table above, participants from group 11 did condition A and reported higher satisfaction ratings. This suggests that when sound effects were included, most people found the button feedback more responsive and entertaining.

On the other hand, group 07 who did condition B produced more inconsistent results with ratings ranging from 2-4. A handful of participants rated their satisfaction with quiet button feedback at 2 or 3. Compared with group 11, the general trend indicates significantly reduced satisfaction, even if some gave the experience a 4.

The difference between the two conditions implies that the sound effect enhances users' experiences of feedback and responsiveness. Overall, participants felt that the interface was more interesting and when sound was available, it gave clearer confirmation of successful button press.

To test whether the difference in satisfaction ratings between the two conditions is statistically significant, we used a non-parametric Mann–Whitney U test because the dependent variable is a Likert scale rating (1–5) and the design is between-subjects, so normality assumptions for parametric tests are not required and may not hold. Using $\alpha = 0.05$, the results showed a statistically significant difference between conditions. $U = 34.0$, $p \approx 0.0098$), with higher satisfaction ratings in the SFX condition. This supports PO2 and PH2.

Both PO2 and PH2 are supported by these findings. Having sound effects improved overall satisfaction and perceived feedback quality, proving that audio cues improve the responsiveness and intuitiveness of button engagement. As a result, this experiment offers compiling evidence that sound effects improve user experience and need to be incorporated into interface design.

Conclusion

In conclusion, our scenario-based fire preparedness game was iteratively improved through usability evaluation and redesign. Experiment 1 showed that changing the main menu button label from “Scenario Info” to “Quick Start” significantly improved user performance and understanding, supporting both P01 and PH1. Experiment 2 found that adding button sound effects significantly increased user satisfaction with button feedback, supporting P02 and PH2. Overall, these findings suggest that clearer action-oriented labels and multi-sensory feedback improve usability and perceived responsiveness in our interface. Future work can strengthen validity by increasing the sample size, testing with a more diverse participant group beyond students, and evaluating learning outcomes in addition to usability perceptions.

Appendix

Appendix A - Experiment 1 Script

Hello, thank you for agreeing to take part in our experiment. Today, we're evaluating how easy it is for users to navigate the main menu of our fire-safety learning game. There are no right or wrong answers — we're testing the interface, not you. Please think aloud if possible, and feel free to ask questions or give us any feedback after the task.

I'm going to show you a simple main menu screen. Your task is to:

Imagine you are returning to this game and you want to continue your last played scenario. Using this interface, navigate to your last play level. You may use any button you think is appropriate. Once I display the screen, I'll start the timer. Just navigate naturally as you would in a real game.

Great, thank you for your participation. Before we end the session, I have one short question: How clear was the purpose of the button you saw? Please rate it on a scale of 1-5: 1 means 'very unclear', and 5 means 'very clear'.

Once again, thank you very much and that completes experiment 1.

Appendix B - Experiment 2 Script

Hello, thank you for agreeing to take part in our experiment. In this task, we're testing your experience interacting with the buttons in our interface. Again, there are no right or wrong answers — we're evaluating the system, not you. You will follow a short set of steps shown on screen. Please complete them as naturally as possible:

1. Quick-start a scenario from the main menu.
2. Return to the main menu.
3. Open the settings menu.
4. Return to the main menu again.

I will start the task when the interface is shown. Just follow the steps exactly once. There is no time limit.

Great, thank you for your participation. Before we end this session, I have one final question: How satisfied were you with the button feedback in this interface? Please rate it from 1 to 5: 1 means 'very dissatisfied', and 5 means 'very satisfied'.

Once again, thank you for participating in our experiment and that's the end of experiment 2.

Appendix C - Experiment 1 Consent

Group 40 representative

PARTICIPANT CONSENT FORM - 1-grp40.pdf

Group 42 representative

PARTICIPANT CONSENT FORM - 1-grp42.pdf

Appendix D - Experiment 2 Consent

Group 11 representative

PARTICIPANT CONSENT FORM -2-grp11.pdf

Group 07 representative

PARTICIPANT CONSENT FORM -2-grp7.pdf

Appendix E - Declare Use of AI

We acknowledge the use of ChatGPT https://chatgpt.com/ to generate
These are the following prompts we used:

- "Are my paragraphs answering the question asked?"
- "Can you help me correct my English for this paragraph?"

We used/modified the output to:

- Change paraphrasing
- Ensure smooth transitions between ideas

Name of Team Members:

Name	Student ID	Signature
KWEK LI XUAN	2402517	lx
SUBHASHINI RAJKUMAR	2402769	SR
MUHAMMAD ADLI BIN JUMAHAT	2401516	Adli
LAI YUEYIN SHYANN	2402456	Shyann
WONG JUN YONG BRIAN	2401610	Brian